

Machine Learning & Scientific Machine Learning

Resit Exam 2024
** WITH ANSWERS **

4 June 2024, 8:30–10:45

You are allowed to write on the exam booklet, and to take it with you afterwards. The exam will be made available on Canvas later today, together with the correct answers. You can copy your final answers onto the booklet, so you can check how you did.

There are 40 questions, worth 1 point each.

To pass the examination, your total points for the exam plus those for the quizzes should be 52 points or more.

Rules:

- You are allowed to use a calculator or graphical calculator.
- You are *not* allowed to use your phone or smartphone.
- The exam is closed-book.
- You are allowed to use the formula sheet provided through Canvas.
- The formula sheet should not have any writing on it, except in the cheat sheet box. Anything inside the box should be written by hand.

Recall questions

1. The most important rule in machine learning is “never judge your performance on the training data.” If we break this rule, what can happen as a consequence?
 - A The loss surface no longer provides an informative gradient.
 - B We get cost imbalance.
 - C We end up choosing a model that overfits. ✓
 - D We commit multiple testing.
2. What is a valid reason to prefer *gradient descent* over *random search*?
 - A My model is easily differentiable. ✓
 - B I need to be sure that I find the global minimum.
 - C My loss function is not smooth.
 - D There is some computation between the output of my model and my loss function, which I do not control.
3. In the slides, we get the advice that “sometimes your loss function should not be the same as your evaluation function.” Why not?
 - A The evaluation function may provide a smooth loss surface.
 - B The loss surface resulting from the evaluation function may have a shape that makes searching for the minimum difficult. ✓
 - C The evaluation function may not be linear, and a loss function must be linear.
 - D The evaluation function may not be a polynomial, and the loss function must be a polynomial.
4. There are many classification problems that are not linearly separable. Which trick often allows us to still separate the classes with a linear classifier?
 - A Adding a regularizer.
 - B Using least-squares-loss instead of log-loss.
 - C Reducing the dimensionality with PCA.
 - D Deriving new features from the existing ones. ✓
5. We are training a classification model by gradient descent, and we want to figure out which learning rate to use, before comparing the model to other classifiers. We try five learning rate values, resulting in five different models. How do we choose among these five models?
 - A We measure the accuracy of each model on the training set.
 - B We measure the accuracy of each model on the validation set. ✓
 - C We measure the accuracy of each model on the test set.
 - D We measure the accuracy of each model on the full dataset.

6. In the naive Bayes classifier, what does the word *naive* refer to?
- A The fact that prior probabilities are not taken into account.
 - B The fact that the resulting classifier does not usually work very well in practice, and is only useful as a baseline.
 - C The fact that the classifier makes an assumption of conditional independence that is not usually true. ✓
 - D The fact that the classifier makes an assumption of independence that is not usually true.
7. The slides mention two ways to adapt a categorical feature for a classifier that only accepts numeric features: integer coding and one-hot coding. Which is **true**?
- A Integer coding always turns one categorical feature into multiple numeric features.
 - B One-hot coding always turns one categorical feature into multiple numeric features. ✓
 - C Integer coding becomes inefficient if there are too many categories.
 - D One-hot coding becomes inefficient if there are too few categories.
8. How does dropout help with the overfitting problem?
- A By propagating the gradient of the loss back down the network.
 - B By randomly disabling nodes in a neural network, to eliminate solutions that require highly specific configurations. ✓
 - C By ensuring that the output distribution of a neural network is normally distributed if the input distribution is.
 - D By converting the scalar backpropagation algorithm to work with tensors.
9. What is the main idea of *instruction tuning*?
- A To finetune the instructions in the dataset on which a large language model is trained. This allows us to train more quickly on less data.
 - B To finetune a large language model on a dataset of natural language instructions. The model will then generalize to types of instructions it hasn't seen before. ✓
 - C In a reinforcement learning setting, to take human instructions in natural language, and to use them to help the agent solve complex planning problems.
 - D In a reinforcement learning setting, to slowly increase the difficulty of the training problems, so that the agent learns to solve simple tasks first, and then progressively more difficult ones.

10. What is the *Markov assumption*? Does the LSTM make the Markov assumption?
- A That the probability of a word depends only on a fixed number of words before it. The LSTM makes a Markov assumption.
 - B That the probability of a word depends only on a fixed number of words before it. The LSTM does not make a Markov assumption. ✓
 - C That it is more likely that a word should be forgotten than that it should be remembered. The LSTM makes a Markov assumption.
 - D That it is more likely that a word should be forgotten than that it should be remembered. The LSTM does not make a Markov assumption.
11. By what method do variational autoencoders avoid mode collapse?
- A By training the “decoder” network through a discriminator.
 - B By using a regularizer to steer the network toward the data average.
 - C By feeding the discriminator network pairs of inputs.
 - D By learning the latent representation of an instance through an “encoder” network. ✓
12. In some machine learning settings it is said that we must make a trade-off between *exploration* and *exploitation*.
- What do we mean by this?
- A That hyperparameter selection (exploration) uses computational resources that can also be used in training the model (exploitation).
 - B That an insufficiently carefully trained model may be biased against minorities.
 - C That a reinforcement learning algorithm needs to balance optimization of its expected reward with exploring to learn more about its environment. ✓
 - D The problem of balancing the loss function with the regularization terms in matrix factorization.

Combination questions

13. You want to search for a model in a discrete model space. Which search method is the **least** applicable?
- A Random search
 - B Simulated annealing
 - C Evolutionary methods
 - D Gradient descent ✓
14. Which is (primarily) a **supervised** machine learning method?
- A Principal Component Analysis
 - B Support Vector Machines ✓
 - C Variational Autoencoders
 - D k-Means clustering

15. Which statement is **true**?
 - A The average error of many models with high bias can be low.
 - B The average error of many models with high variance can be low. ✓
 - C A model with high bias must have low variance.
 - D High bias is an indication of overfitting.
16. Which metric can you **not** compute from the confusion matrix?
 - A Accuracy
 - B True positive rate
 - C Recall
 - D Area under the curve ✓
17. Why is the ReLU activation function often preferred over the sigmoid activation function, for hidden nodes?
 - A It causes more vanishing gradients, which help learning.
 - B The sigmoid function cannot be used with gradient descent.
 - C Its derivative is almost always either 0 or 1, reducing vanishing gradients. ✓
 - D The sigmoid function contains a point where the gradient is not defined.
18. One piece of advice we give in the course is that *the mean is sensitive to outliers*. What does this mean, and why is it the case?
 - A It means one instance can affect the mean much more than the others if its value is unusually large. This happens because the mean minimizes the absolute magnitude of the distances to all instances.
 - B It means one instance can affect the mean much more than the others if its value is unusually large. This happens because the mean minimizes the squares of the distances to all instances. ✓
 - C It means it can be a poor representation of the data if all instances are far from the average. This happens because the mean minimizes the absolute magnitude of the distances to all instances.
 - D It means it can be a poor representation of the data if all instances are far from the average. This happens because the mean minimizes the squares of the distances to all instances.
19. To make the backpropagation algorithm more broadly applicable, we introduced the *multivariate chain rule*. What was the main reason for this?
 - A To allow us to use backpropagation on functions that are expressed in terms of computation graphs.
 - B To allow us to apply backpropagation to computation graphs where the output depends on the input along multiple paths. ✓
 - C To allow us to apply backpropagation to computation graphs where the output depends on the input only along a single path.
 - D To make backpropagation a fully symbolic algorithm instead of a mixture of symbolic and numeric computation.

20. In a maximum-margin loss, which is **not** an advantage of using *soft margins*?
- A It allows some points to fall inside the margin.
 - B It allows some points to be misclassified.
 - C It allows us to use gradient descent for optimization, which a hard margin doesn't. ✓
 - D It finds better decision boundaries even for linearly-separable classes.
21. The Variational Autoencoder (VAE) differs from a regular autoencoder in several aspects. Which is **not** one of them?
- A It includes a discriminator, which tries to tell the difference between data points and samples from the generator. ✓
 - B It has an added loss term that ensures that the data looks like a standard normal distribution in the latent space.
 - C For a given instance, the encoder produces a distribution on the latent space, instead of a single point.
 - D It includes a sampling step in the middle, where a latent vector is sampled from the distribution provided by the encoder.
22. Sometimes we want to learn a model that maps some input to some output, but in some aspects we also want the model to behave like a generator. For instance, if we train a model to colorize photographs, we don't want it to be purely deterministic: to colorize the label of a beer bottle, it should randomly imagine some colors even if it can't infer the correct colors from the input. Which GAN approach is designed to accomplish this?
- A Vanilla GAN
 - B Conditional GAN ✓
 - C CycleGAN
 - D StyleGAN
23. When we say that self-attention combines the best properties of RNNs and Convolutions, what do we mean?
- A Like an RNN, it can look at any point in the input to compute the current output, and like a Convolution, each output can be computed in parallel. ✓
 - B Like a Convolution, it can look at any point in the input to compute the current output, and like an RNN, each output can be computed in parallel.
 - C Like a Convolution, it's an entirely linear operation, and like an RNN it can theoretically work on an arbitrarily long sequence with a finite memory.
 - D Like an RNN, it's an entirely linear operation, and like a Convolution it can theoretically work on an arbitrarily long sequence with a finite memory.

24. The standard decision tree algorithm doesn't stop adding nodes until all leaves either contain no data instances, or only instances with the same label (or all features have been used). Why is this a problem, and what is the default solution (mentioned in the slides)?
- A It's a problem because the algorithm may never terminate. To solve it, we can use a validation set to see if removing nodes improves performance.
 - B It's a problem because the algorithm may never terminate. To solve it, we can remove features from the data, so that fewer splits are available.
 - C It's a problem because we may be overfitting on the training set. To solve it, we can use a validation set to see if removing nodes improves performance. ✓
 - D It's a problem because we may be overfitting on the training set. To solve it, we can add features to the data that are computed from the original features.
25. What problem, if it exists for a single model, **cannot** be solved by training an ensemble of those models?
- A High training time. ✓
 - B High bias.
 - C High variance.
 - D High overfitting.

Application questions

We have the following training set:

	x_1	x_2	label
a	2	5	Pos
b	4	2	Pos
c	3	1	Pos
d	1	4	Neg
e	5	5	Pos
f	0	6	Pos
g	-3	-7	Neg

For the following questions, it helps to draw the data and the classification boundary in feature space.

We use a linear classifier defined by

$$c(x_1, x_2) = \begin{cases} \text{Pos} & \text{if } x_1 - 0 \cdot x_2 > 3 \\ \text{Neg} & \text{otherwise.} \end{cases}$$

26. If we turn **c** into a *ranking* classifier, how does it rank the points, from most **Negative** to most **Positive**?

- A f a b e g a c
- B g f d a c b e ✓
- C e b c a d f g
- D g d c f a b e

27. How many ranking errors does the classifier make?

- A 1 ✓
- B 2
- C 3
- D 4

If you answered 3, make sure you understand the difference between a *ranking error* and a *classification error*. Classification errors are instances that the classifier classifies incorrectly (in this case the instances a, c, and f) and ranking errors are *pairs* of **positive** and **negative** instances that a ranking classifier ranks the wrong way around. In this case (f, d).

28. If we draw a coverage matrix (as done in the slides), what proportion of the cells will be red?

- A $\frac{1}{10}$ ✓
- B $\frac{1}{6}$
- C $\frac{1}{4}$
- D $\frac{1}{3}$

The size of the coverage matrix is the number of positives times the number of negatives ($4 \times 3 = 12$). The number of red cells is the number of ranking errors (1) so the proportion of red cells is $1/12$.

We will use the backpropagation algorithm to find the derivative of the function

$$f(x) = 9x^2(5x - 2)^2$$

with respect to x .

First, we break the function up into modules:

$$a = 5x - 2$$

$$b = 3x^2$$

$$c = 3a^2$$

$$f = bc$$

29. What should be in the place of the dots?

- A $c = 3a^2$, $b = 3x^2$ ✓
- B $c = 3a^2$, $b = x^2$
- C $c = 3a$, $b = 3x^2$
- D $c = 3a$, $b = x^3$

30. In terms of the *local* derivatives. Which is the correct expression for the derivative $\frac{\partial f}{\partial x}$?

- A $6cx + 30ab$ ✓
- B $18c + 30ab$
- C $18cx + 90ab$
- D $6c + 30ab$

Here are two distributions, p and q , on the members of a set $X = \{a, b, c, d\}$. We will use binary entropy (i.e. computed with base-2 logarithms). Note that in the computation of the entropy $0 \cdot \log_2(0) = 0$, but otherwise, $\log_2 0$ is undefined as usual.

	p	q
a	$\frac{4}{16}$	$\frac{4}{16}$
b	$\frac{2}{16}$	$\frac{8}{16}$
c	$\frac{2}{16}$	$\frac{4}{16}$
d	$\frac{8}{16}$	0

31. What are their entropies?

- A $H(p) = 2.25, H(q) = 1.25$
- B $H(p) = 2.25, H(q) = 1.50$
- C $H(p) = 1.75, H(q) = 1.50$ ✓
- D $H(p) = 1.75, H(q) = 1.25$

32. What are their cross-entropies?

- A $H(p, q)$ is undefined, $H(q, p)$ is undefined.
- B $H(p, q) = 0.875, H(q, p) = 2.75$
- C $H(p, q)$ is undefined, $H(q, p) = 2.75$. ✓
- D $H(p, q) = 0.875, H(q, p)$ is undefined.

We are given a dataset of email, labeled **spam** and **ham**. The total number of words in the **spam** and **ham** datasets is **10 000** and **100 000**, respectively.

On this dataset, we want to train a *first-order Markov model* as a basis for a Bayes classifier to detect spam.

The following table shows the frequency of several bigrams and uni-grams in the dataset.

uni- and bigrams	frequency	
	spam	ham
pay	2 000	10 000
your	1 000	3 600
invoice	5 000	2 400
today	1 000	2 000
pay your	50	600
your invoice	50	120
invoice today	50	120

We have two *priors* on how likely an email is to be spam. **Prior 1** says that an email is as likely to be **spam** as **ham**. **Prior 2** says that the probability that a given email is **spam** is 90%.

Consider the following email:

email: pay your invoice today

33. How is the email classified?

- A With both priors as **ham**.
- B With both priors as **spam**.
- C Under prior 1 as **spam**, under prior 2 as **ham**.
- D Under prior 1 as **ham**, under prior 2 as **spam**. ✓

$$\begin{aligned}
 p(\text{pay, your, invoice, today} \mid \text{spam}) &= p(\text{today} \mid \text{invoice, spam}) \cdot \frac{50}{5\,000} = \frac{1}{100} \\
 &\quad p(\text{invoice} \mid \text{your, spam}) \cdot \frac{50}{1\,000} = \frac{1}{20} \\
 &\quad p(\text{your} \mid \text{pay, spam}) \cdot \frac{50}{2\,000} = \frac{1}{40} \\
 &\quad p(\text{pay} \mid \text{spam}) \cdot \frac{2\,000}{10\,000} = \frac{1}{5}
 \end{aligned}$$

$$= \frac{1}{100} \frac{1}{20} \frac{1}{40} \frac{1}{5} = \frac{1}{2 \cdot 4 \cdot 5} \cdot \frac{1}{100 \cdot 10 \cdot 10} = \frac{1}{4} \cdot 10^{-5}$$

$$\begin{aligned}
 p(\text{pay, your, invoice, today} \mid \text{ham}) &= p(\text{today} \mid \text{invoice, ham}) \cdot \frac{120}{2\,400} = \frac{1}{20} \\
 &\quad p(\text{invoice} \mid \text{your, ham}) \cdot \frac{120}{3\,600} = \frac{1}{30} \\
 &\quad p(\text{your} \mid \text{pay, ham}) \cdot \frac{600}{10\,000} = \frac{3}{50} \\
 &\quad p(\text{pay} \mid \text{ham}) \cdot \frac{10\,000}{100\,000} = \frac{1}{10}
 \end{aligned}$$

$$= \frac{1}{20} \frac{1}{30} \frac{3}{50} \frac{1}{10} = \frac{3}{2 \cdot 3 \cdot 5} \cdot \frac{1}{10 \cdot 10 \cdot 10 \cdot 10} = \frac{1}{10} \cdot 10^{-4} = 10^{-5}$$

To get the classification we multiply both by their prior and check which is higher. Under **prior 1**, both prior probabilities are equal so the classification is **ham** ($10^{-5} > \frac{1}{4}10^{-5}$).

Under prior 2, we get

$$p(\text{spam} | E) \propto \frac{1}{40} \cdot 10^{-4} \cdot \frac{9}{10} = \frac{9}{40} \cdot 10^{-5}$$

and

$$p(\text{ham} | E) \propto 10^{-5} \cdot \frac{1}{10} = \frac{4}{40} \cdot 10^{-5}$$

so the classification is **spam** (because $4/40 < 9/40$).

34. Under prior 1, what is the probability that the email is **spam**?

- A $\frac{1}{8}$ B $\frac{7}{8}$ C $\frac{1}{5}$ ✓ D $\frac{4}{5}$

35. Under prior 2, what is the *probability ratio* of **ham** over **spam** for the email?

- A $\frac{7}{8}$ B $\frac{8}{7}$ C $\frac{4}{9}$ ✓ D $\frac{9}{4}$

Consider the following task. The aim is to predict the class y from the binary features x_1 , x_2 , x_3 and x_4 .

x_1	x_2	x_3	x_4	y
B	A	B	B	No
A	B	A	B	Yes
A	A	A	A	No
A	A	B	B	No
B	B	B	A	Yes
A	B	B	B	Yes
A	A	A	A	No
B	B	A	A	No
A	A	A	B	Yes
B	B	B	A	Yes
B	A	B	A	No
B	B	A	B	Yes

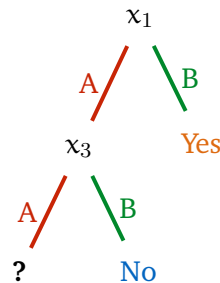
36. In standard decision tree learning (as explained in the lectures), without pruning. Which would be the first feature chosen for a split?

- A x_1 B x_2 ✓ C x_3 D x_4

37. If we *remove* that feature from the data, which would be chosen instead?

- A x_1 B x_2 C x_3 D x_4 ✓

With the same data, consider the following (partial) decision tree:



Note that this is just an arbitrary tree, not necessarily corresponding to the answers in the previous questions.

We can place either x_2 or x_4 on the open node (indicated by the question mark).

38. At this point, what is the information gain for x_4 ?

A -1 B 0 C 0.5 D 1 ✓

Consider the following function:

$$\mathbf{y} = 2\mathbf{x}^{\frac{1}{2}} + \mathbf{b}$$

Here $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{b}$ are vectors and the square root is applied element-wise (that is, $\mathbf{x}^{\frac{1}{2}}$ is a vector whose i -th element is $x_i^{\frac{1}{2}}$).

This function is part of a larger computation graph, resulting in a scalar loss l . We want to implement this computation in an automatic differentiation (AD) system, as discussed in the lectures.

39. Work out the scalar derivative of y_i over x_j . Which is the correct solution?

A $\frac{\partial y_i}{\partial x_j} = \begin{cases} 2/\sqrt{x_i} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$

B $\frac{\partial y_i}{\partial x_j} = \begin{cases} 2/\sqrt{x_i} & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases}$

C $\frac{\partial y_i}{\partial x_j} = \begin{cases} 1/\sqrt{x_i} & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$ ✓

D $\frac{\partial y_i}{\partial x_j} = \begin{cases} 1/\sqrt{x_i} & \text{if } i \neq j \\ 0 & \text{otherwise} \end{cases}$

In scalar terms, the computation is $y_i = 2x_i^{\frac{1}{2}} + b_i$. When we work out $\frac{\partial y_i}{\partial x_j}$, we can tell immediately that y_i only contains x_j if $i = j$, so the derivative is

0 otherwise (leaving A and C as answers). If $i = j$, the the question reduces to

$$\frac{\partial y_i}{\partial x_i} = \frac{\partial 2x_i^{\frac{1}{2}} + b_i}{\partial x_i} = 1 \cdot x_i^{-\frac{1}{2}} = \frac{1}{\sqrt{x_i}}.$$

The AD system will give us a vector $\mathbf{y}^\nabla$, such that $y_i^\nabla = \frac{\partial l}{\partial y_i}$. Using $\mathbf{y}^\nabla$, we want to efficiently compute a vector $\mathbf{x}^\nabla$ such that $x_i^\nabla = \frac{\partial l}{\partial x_i}$. Let $\otimes$ represent element-wise multiplication of vectors, and let all exponents be element-wise as before.

40. Which operation computes this vector for us?

- A $\mathbf{x}^\nabla = \mathbf{y}^\nabla \otimes \mathbf{x}^\nabla$
- B $\mathbf{x}^\nabla = \mathbf{y}^\nabla \otimes \mathbf{x}^{-\frac{1}{2}}$ ✓
- C $\mathbf{x}^\nabla = \mathbf{y}^\nabla \otimes \mathbf{x}^{\nabla \frac{1}{2}} + \mathbf{b}$
- D $\mathbf{x}^\nabla = \mathbf{y}^\nabla \otimes \mathbf{x}^{-\frac{1}{2}} + \mathbf{b}$

First, we work out x_i^∇ .

$$x_i^\nabla = \frac{\partial l}{\partial x_i} = \sum_{\mathbf{k}} \frac{\partial l}{\partial y_{\mathbf{k}}} \frac{\partial y_{\mathbf{k}}}{\partial x_i} = \sum_{\mathbf{k}} y_{\mathbf{k}}^\nabla \frac{\partial y_{\mathbf{k}}}{\partial x_i}.$$

In the factor on the right, $y_{\mathbf{k}}$ only contains x_i if $\mathbf{k} = i$. All other terms of the sum are zero, so we get

$$x_i^\nabla = y_i^\nabla \frac{\partial y_i^\nabla}{\partial x_i} = y_i^\nabla \times x_i^{-\frac{1}{2}}$$

where the last step fills in the answer to the previous question. This tells us that we are looking for a vector $\mathbf{x}^\nabla$ whose i -th element is given by $y_i^\nabla \times x_i^{-\frac{1}{2}}$. We can compute such a vector as $\mathbf{x}^\nabla = \mathbf{y}^\nabla \otimes \mathbf{x}^{-\frac{1}{2}}$.

Thank you for your effort. Please check that you've put **your name and student number** on the answer sheet.